MNRI Proprioceptive Cognitive Reflex Integration

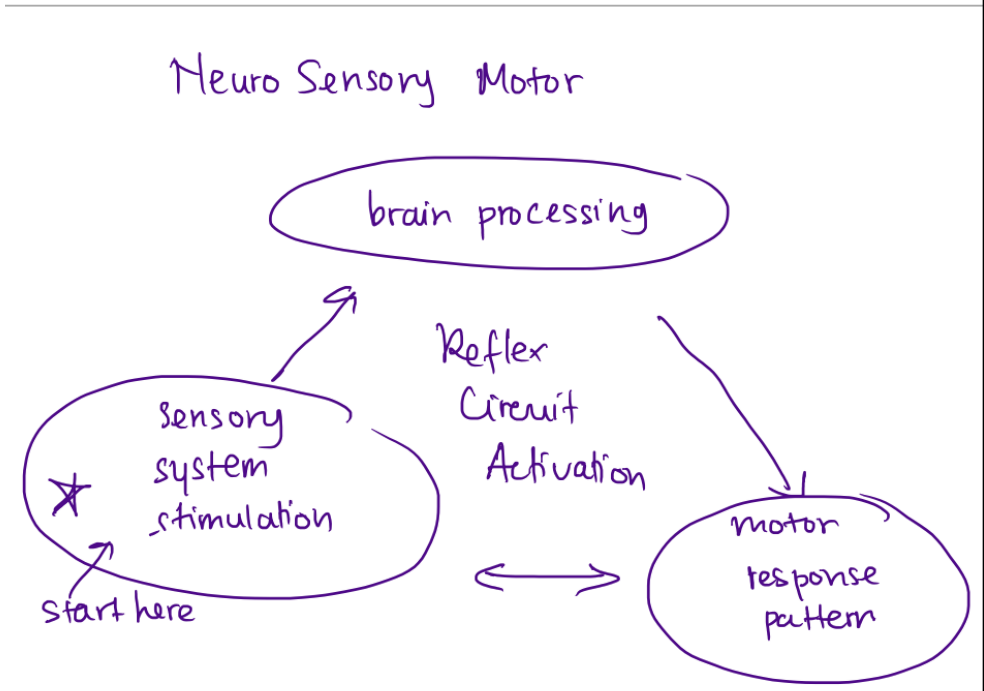
Personal Notes: Gaby Jonatan

Lecture by: Theresa Fischer

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Introduction

- Precursor of MNRI method (additional to previous notes)
  - Bernstein
  - Charles Sherrington (coined proprioceptive
- MNRI principles: simple is profound, the missing link, connect sensory stimulation with specific motor response (in sensory integration therapy, you give the input, but miss the output) – unless you get the vestibular system and the proprioceptive to the brain, you will not get the change. Always go for safety (most of them live in the fight/flight). When we put our hands on someone: clear mind, grounded, be present with our hands.



Introduction continue

- What is a reflex? A unit of nervous system functioning, unit for survival and protection, unit of genetic motor memory, unit for neuro-development and positive compensation (archetype -> reflex-> higher level motor skills), bridge between automatic responses and consciously learned movements and skills, serves as neurophysiological foundation for cognitive development, serve as a language to reach motor responses and memory.
- “The first infant movements do not disappear; they continue to work in union with higher nervous formation, entering into their structure as the subordinated instances and transferring part of their functions to higher, “younger” and new centers” – lev Vygotsky.
- These reflexes don’t go away, stay in the brain stem.
- In MNRI, we integrate (myelination - protein sheet on the nerve). We don’t want them to drive our movements, we want skilled movements, but we need the reflexes to be properly integration. In bobath system, we inhibit the reflex but the moment they are on alert, the reflex will come back out because they are not integrated.

Introduction continue

- MNRI indications: autism, asperger syndrome, ADHD, ADD, fear/phobia, aggression, CP,
- Nerve Net Systems:
  - The brain and nervous system contain billions of interconnected neurons
  - The number and organization influence everything we do
  - At birth infants have all, or most of the brain cells they need.
  - Wiring and connections between cells need to be built. (The neurons are all there but not connected)
  - By 3 months old, the synapses multiply more than 20 times. It is important for the baby to feel safe and receive rich sensory input.
  - Early **movement** and **sensory experiences** create synapses. In stress trauma, the synapses burn out (fried out).
  - Repetition strengthens synapses
  - Experiences create brain codes and patterns, patterns develop sense of “who I am” (100 billion nerve cells, 100 trillion synapses, dozens of neurotransmitters).

Proprioceptive/Cognitive Integration

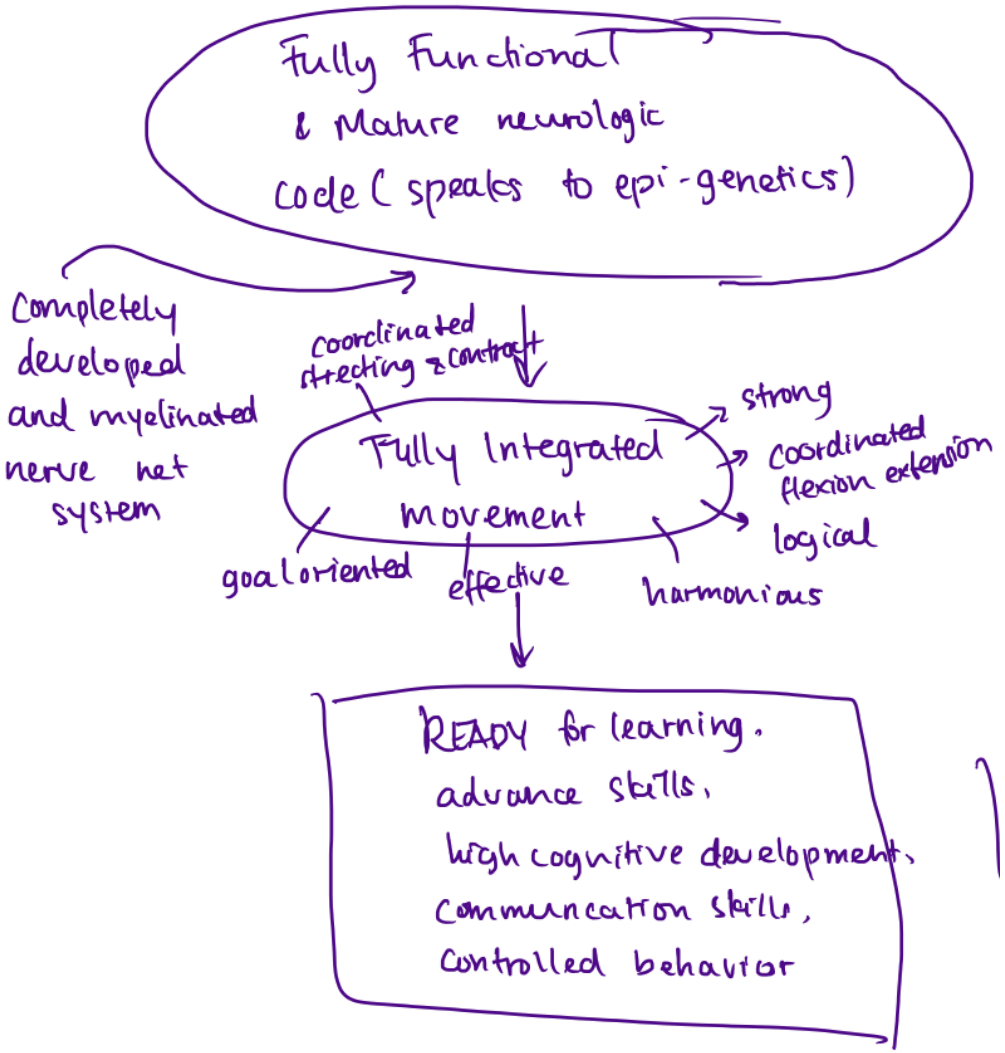
- Many great innovators of education agree movement is one of most important elements:
  - Motor development
  - Cognitive development
  - Emotional development
- Higher brain function such as literacy are dependent on success of development of lower brain function.
- Without development of lower brain children learn but with much more effort… challenged learners.
- Motor activity and proprioception function as a single unit.
- This unit created by nature to assist brain development and function.
- At birth proprioceptive system is explored in environment that includes gravity (we need proper vestibular and proprioceptive at birth).
- Proprioception is developed along with conditioned (variant, learned, from environment and gravity) and unconditioned (basic pattern in utero) motor reflexes.

Proprioceptive/cognitive integration

- Proprioception organizes:
  - Gravity and antigravity reflexes (starts at 25 days in utero when spinal cord developing)
  - Grounding reflex
  - Stability reflex (horizontal line)
  - Balance reflex
  - Head righting reflex

Proprioceptive/Vestibular system

- Develops in utero
- Big squeeze gives biomedical regulation to set up for gravity (coming down the vaginal canal).
- 5 reflexes interact with gravity and organize proprioceptive/vestibular systems.
- Semicircular canals are developed in aquatic environment
- Matured through birthing, gravity and biomedical interactions.


Fully Functional
& Mature neurologic
code (speaks to epi-genetics)
Completely developed and myelinated nerve net system
coordinated stretching & contract
Fully Integrated movement
strong
coordinated flexion extension
logical
harmonious
goal oriented
effective
READY for learning.
advance skills.
high cognitive development.
communication skills,
controlled behavior

Purpose of this class

- Link the role of proprioceptive/vestibular system to cognitive systems
- Role of proprioceptive-vestibular system in gravity, grounding and stability for cognitive functions
- Conscious and unconscious postural and motor control
- Link between coordination, development of inner locus (emotional) control

- “Attendance at school should not inhibit learning” – Mark Twain

Four Major parts of the brain

- Cortex
- Cerebellum
- Diencephalon
  - Thalamus (sensory gateway to the brain)
  - Hypothalamus
- Brain stem
  - Midbrain
  - Pons
  - Medulla oblongata

Movement



- Essential for learning
- Integrates and anchors new information
- Integrates experience in our neural networks
- In utero gives first sense of the world
- Begins knowledge of laws of gravity
- Build on movement to shape our vision, to explore our environment
- Forms our ability to interact with people around us

Movement

- “Ready to learn”
  - Immature neurological system
    - Unable to sit
    - Difficult following directions
    - Unable to horizontally scan
    - Poor fine motor (lack of speech, when they don’t get to play, they lack language)
    - Clumsy
    - Behavioral challenges (aggression, anger issue)

Rules of movement

- Breathing + stability + alignment + muscle tone + joint freedom (archetype) = MOVEMENT
  - Our archetype movements give us our freedom of movement
  - Take breath away, movement is forced
  - Stability away, cannot stand up staight
  - Joint freedom away, stuck (CP kids).
- Movement Anchors THOUGHT
  - We learn better when we are moving

Movement

- Following work of Piaget and Vygotsky, Dr Masgutova, Dennison with brain gym, jean Ayres saw movement and reflex integration as a foundation to neural development
- Learning/literacy dependent on lower brain functions
- Humans described as having 5 senses (vision, auditory, smell, touch, taste)
- Form complete picture requires integration of all
- Dr Sherrington described proprioception as the 6th sense
- Proprioception is body's sense of itself in space (no body awareness, don't know where they are in space).
- Proprioception allows us to explore environment (kids who use the wall, kids who cling to wall – they don't have proprioception, they use mom or wall to maneuver the environment.
- Proprioception gives feedback: optimal muscle contraction to **be centered**.
- When stress interferes with balance of this system, we become **uncentered**: lose balance and physical sense of self in space. (Just doing for the sake of doing, driving to work when it is a holiday, some kids function that way every time)

Movement

- In utero development includes major brain systems of tactile, auditory, vision and motor.
  - These systems develop in relationship to vestibular system
    - Gives us our model of 3-dimensional space
- Our 3-d world gives us framework for which all other sensory data must be integrated (I don't know where I am, and I don't know where that sound came from so I don't know where to go from here)
- Successful integration is necessary for brain functions
  - Reading
  - Writing
  - Academic achievements
- Observing a child move is indirectly observing efficiency of child's brain processing (is it coordinated, is it logical, are they able to walk from a-b —> check their level of integration
- Smooth/coordinated movements = integration
- Rigid/uncoordinated movements = faulty integration
- This is manifestation in the learners problems: poor academic achievement and a struggling learners.

Movement

- Greater academic achievement good in theory only (push academic in young age – another area will suffer)
- Negative consequences in our students' learning
- Lack of brain-body work development
  - Motor planning (jumping, bouncing, running)
  - Auditory (retaining letter, sounds, listening to teacher). We start with ATNR (one ear at a time), we then work in midline then both ears are activated. We say look to the front (they look front but lost their hearing).
  - Sensory processing (behavior, focus, attention) – they are just busy keeping their body up right, so they usually the class clown.
  - Organizational skills used for math (if you cannot jump, you cannot make vertical letters, if you cannot go side to side, you cannot do math, if you cannot do cross crawl). The way we move affects the way we learn. We need to give them find their midline, ground them, to be able to stabilize themselves.

Proprioception

- Proprioception is developing in utero
- Proprioception and motor activity function as one unit
- Proprioceptive links “body-brain” system
- “Sense of self”
- Awareness of body position
- Awareness of rate of movement
- Awareness of strength being used for a movement
- Proprioception senses internal state of body (tactile system tells us about the external part of the body).
- Connection between brain and muscles
- Receptors inside body “interoceptive sense”
- Senses perceived from:
    - Posture
    - Specific body motions
    - Pain of internal organs
- basis for unconscious responses and conscious movements
- Allows the mind to understand the body without vision (eyes closed but able to tell where their hand is located whether or not their elbow is bent)

Function of proprioception (p.17)

- Survival: keeps musculoskeletal system free from injury (low tone kids are very anxious, they think they are hurt so they stop moving).
- Development: stores new experiences used for learning new skills and goal oriented movements. It stores motor memory so it can store experiences.
- Organizes **Postural Control** and smooth controlled movement
  - Vestibular system and proprioception develop together in utero. So we work them together in the MNRI.

2 levels of proprioception (p.18)

- Conscious sensation
  - Utilized to facilitate complex motor activity
  - Ability to think, plan, and execute a task
  - Utilized when there is confusion in a movement (when we don't know a move – summersault , we stop and we use proprioception to execute that move).
- Unconscious Sensation
  - Coordinates basic posture during sitting, standing and simple gait
  - Provides connection between brain and muscles
  - Produces smooth, spontaneous and controlled movement
    - This is what we use when we drive a car: we don't have to look at your foot to do the gas and the break.

Proprioception

- Proprioception is power!
  - This power is locked in the unconscious mind
  - Strength, power, balance, stability, movement are directed unconsciously.

Types of proprioceptors (p.17)

- 1. Muscle spindles (belly of the muscle)
  - Found in skeletal muscle
  - Monitor muscle length (the contraction and expansion of muscle)
- 2. Golgi Tendon Organs
  - Found in tendon at muscle/tendon junction (between muscle and bone)
  - Monitors muscle tension (tight hamstring)
- 3. Joint kinesthetic receptors: pacinian/Rufini corpuscles
  - Found in capsule/ligament of synovial joints (joint lubrication)
  - Monitors joint angle and capsule pressure
  - Responds to pressure and acceleration/deceleration of joints during movement, tells brain how fast the muscle is moving.
- 4. Sensory hair cells (this is in vestibular system – in proprioceptors)
  - Located in inner ear
  - Respond to change in head position.

Proprioceptive and vestibular system

- United systems
- First job of prop system to develop anti-gravity and this is done with reflex patterns. (Proprioceptive input: hand inside palm, to get the grasp reflex).
- Every vestibular reflex activated by proprioceptive action (your headrighting, balance). The movement enlist the balance reflex.

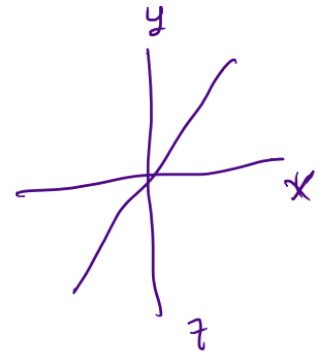
- Vestibular reflexes
  - Equilibrium /balance
  - Trunk extension
  - Galant
  - Balancing
  - Grounding
  - Stability
  - Head righting
  - Vestibular ocular (ocular-vestibular)
  - Kinetic optical (Opto-kinetic)

- Proprioceptive reflexes
  - Trunk extension
  - Galant
  - Balancing
  - Grounding
  - Stability
  - Head righting
  - Infant stretch - when infant do full body stretch and takes in air (you want to see baby do this 17 times a day)

- All spinal reflexes are Both Vestibular and Proprioceptive Reflexes

Dr. Bernstein (1896-1966) the Father of biomechanics and physiology of movements

- Bernstein Principles:
  - 6 degrees of freedom
    - Life happens in 3 dimensions
    - We have the 3 dimensions/axis within our body.
    - Each dimension has 1 degree of linear movement and 1 degree of rotational movement.
    - All 3 dimensions cross at our center of gravity.
- 3 motor coordination systems: goal of every reflex is to be in balance with gravity.
  - Starting from front/back, upper/lower, right/left



Linear:
x: right / left
y: up / down
z: front / back

rotation:
x: no z: maybe
y: yes

Gravity

- Natural force that works through center of gravity to earth.
- Gravity shapes life! — we have to cooperates with gravity because if we don’t, gravity always wins. We will bend our body if we don’t cooperates with gravity.
- Affects body structure and posture
- Force/wave picked up by vestibular system: where I am in space? – it is the vestibular system that picks up gravity and tells us where we are.
- Infants have unconditioned basic reflex patterns in utero (p.35)
- Infants introduced to gravity at birth
- Vestibular system developed at birth but not practiced in gravity. Body is not dependent on it in utero.

Gravity reflex (p.35)

- Brain and body response to force of gravity result of:
  - Body structure – if you have bad posture, you don't take on gravity well so you will bend.
  - Body position in space
  - Movement
- Gravity reflex triggered:
  - Loss if coherency of gravity and body midline

Gravity line

- Critical for development of antigravity and postural control
- Critical for development of sensory systems
- Strong relationship with horizontal / stability lines
- Want strong median gravity line for development of corpus callosum
- Formation of gravity line directly influences development
  - Dimension of laterality: right/left
  - Dimension of focus: front/back
- Right/left development
  - Strengthening of corpus callosum
  - Right and left hemisphere communication
  - Analytical and gestalt thinking
- Front/back development (hans selye)
  - Green light: backward causes hyperactivity (inability to
  - Red light: forward decreases abstract thinking

Front/back MCS

- Focus (When we move forward – the ability to focus, when we go back: ability to look the big picture).
- Integration of brainstem and cortex
- Attention and comprehension. (ADHD kids – test their gravity line).
- Position in space.
- Movement patterns originating in spine and postural alignment. (Originated in the spine).
- Expressive language skills: speaking and writing. Being able to stand increases their ability to express themselves.
- Choose to participate or withdraw: tendon guard.

Right/left MCS

- The highest, the most complex. They communicate between right and left brain: corpus callosum
- Laterality dimension
- Integration of right-left hemisphere (right: analytical, left: gestalt. Right is going to give us C-A-T, left is going to give us CAT).
- Homolateral movement
- Binocular visiton
- Binaural hearing
- Processing and communication

Group activity

- Assess symmetry of horizontal lines of partner
  - Sphenoid
  - Eyes
  - Hyoid
  - Clavicles
  - Shoulders
  - Hips
  - Knees – top and bottom of the platela, whether you hyper extend (vestibular shut down).
  - Ankles – ankle bones and if the head is in line
  - Foot – flat, turned out, or
  - Head midline
  - Feel weight on R/L
  - Feel weight on front back foot
  - Where does gravity line fall
  - Middle finger – when they are standing of the leg.
  - What direction are the eyes when they are standing

Gravity line/shock absorption

- If we have good gravity line - we can move like hula hoop
- Bad gravity line: rounded shoulder (because of cell phones), dysfunctional alignment (ankle not balanced, outward one side), look at the stance, if wide stance they are looking for stability. Foot out – more stability.
- Body forward: lack of abstract thinking and imagination
- Body back: fight/flight

1. Antigravity and shock absorption activation

- Absorption for feet
  - A. Move side to side, you hold the partner on side of shoulder and help them move side to side: move right, squeeze and then middle. Left, squeeze and then middle. Move 7 to 11 degrees to the side. Only move slightly. Teach them boundaries, only to the point they move sideways and can be let go and they still stands.
    - We do right and left
  - B. Get a tape that is eye level and put a arms length width on a wall, put a balloon vision in the middle of the tape, a little above the eye. (A lot of children cannot move side to side and maintain vision focus – gaze stabilization). Make sure shoulder are level. Hand position: thumb behind shoulder, 1 finger on top of shoulder, the rest behind it.
    - Move right, hold for 7 counts, and middle.
  - C. When we have a moving object (3 dimensional object: yellow, orange, or green).
    - Body move right, balloon down right above chest
    - Body move right, balloon move up above head
    - Body move left, balloon move down
    - Body move left, ballon move up. Always end at balloon going up.

1. Antigravity and shock absorption

- D. Absorption of foot arch
  - Top of hamstring, stroke down to knee, flex knee and stabilize.
  - On adult, use body put knee on shoulder, use both hands together (on children, just use hand) and traction outward toward toe
  - Lift slightly the leg from mat, and traction femur up and down
  - Ankle traction
  - Foot flat, bend toes, stabilize the leg, lift up on toe and heel, push down on arch. Hand hold on toes is slightly just to get a flat, not bending the toes. Other hand would be on heel to lift up, and 2/3 fingers on the arch.

2. Gravity line activation in motor stages

- A. Lying down supine (on back). Gravity line keeps us vertical. But when we were born we don't have that, so we practice gravity line until we can come up right on two feet. Kids who are not on the floor, they cannot practice gravity line. For babies and adults who cannot stand up, we start with them lying on back.
  - Put own hand on middle of ribs, and another hand above ball of hip.
  - Upper hand hold towards opposite shoulder, and other hand going into middle. Find a parallel line through the middle. Make sure the body stays straight.
  - Add visual aid above eye, and see if it is harder or easier
- B. Lying prone (on tummy) and supporting elbow.
  - Give hand supporting on wrist accent
  - Give accent into the forearm (when we push elbow down, body comes up).
  - Put hand inside elbow and give pressure out. (Patient will feel have to go up). Hold 7 counts. On Kids with low tone, give support outside elbow to teach the body.
  - Go inward to hold 7 counts.
  - Shoulder blades (little hugs with fingers) towards down, so you can pull up the head (direction of tonic labirinthine).
  - On kids who can understand, you say: push your elbow down (you'll see head go up).

2. Gravity Line activation in motor stage

- C1. In STNR on Knees
  - Hold inside of upper arm
  - Hold outside upper arm
  - Do shoulder hug
  - Do rib and hip on opposite body, and it'll activate midline. To straighten the whole midline.
- C2. STNR on all fours
  - Hold inside upper arm
  - Hold outside upper arm
  - Go on scapula and hold down and hold up.
  - Go on hip, hold down and hold up. Trunk extension
  - Opposite rib and hip straightening midline
- D. Sitting On criss cross
  - Opposite rib and hip straightening midline.
- D2 sitting on chair
  - Hands on knees and bring elbow straight
  - Hold inside upper arm, hold outside upper arm
  - Do rib and hip opposite body to straighten midline
  - Hold and quick release and the body will shock a bit and come back to midline.

3. Gravity Line Activation Within Comfort Zone (day2)

- A. Slowly incline side to side, front and back
- B. With a tape on a wall, a bit above the partner's eye level. The wide can be arms width or longer. Ask them focus on the line. Move them slightly right, left, front, and back. Use the tape to get eye leveling (anyone who has a head tilt or head sideways, ask them to look at the tape).
  - Living in the city, we don't look at the horizon anymore in nature.
  - For toe walkers, put the tape on the wall, to get them to eye level and bring their body down.
- With a balloon on the tape. Balloon starts in the middle.
  - Body leans right, balloon goes right until end of the tape
  - Body leans left, balloon goes left until end of tape.
  - Body leans forward and balloon goes right, body leans left balloon goes left.
  - We move with them, arms bend when holding the partner's shoulder. (We don't want to see the body turning).
- For Active, we add a hold. Move right, hold with one hand, palm open. Front: move in front of them (out of line of vision) and hold from front

4. Gravity Line Activation To Widen the Comfort Zone

- In the book, we hold with one hand
- Tips: hold on both shoulder and hip. Hold the hip to get them move from comfort zone faster.
- Ask to hold, ask to lean more, and then hold again. And then let go of the arm that was holding and put slight touch on the opposite shoulder.
  - First no balloon
  - Balloon stationery
  - Balloon move opposite direction
  - Balloon goes on excursion and not stopping in the middle.
- If partner falling to front, hold sternum and core for them to hold together
- We want slight tension on the core – breathing exercise: inhale nose, exhale through teeth —> your belly engage slightly.

Body Mass Activation and Vertical Visual Perspective Activation

- On the wall, back to the wall.
- Stroke down the knee, in-hook-down and traction down (hold). The gravity line will go the arch.
- Remind the body that you are one unit. Stroke from the scapula down to S2.
- You can also do this lying down. Trunk extension standing with elbow bent.
- On hyper-extended knee, take out the hyper extension - the partner's body might fall on you (make sure you are sitting/or on a bolster). When this happens, you can do the body midline hold on rib and opposite hip. If this is still not there, go back to lying down trunk extension: S2 + heel extension.

Corpus Callosum

- Connection/communication of 2 hemisphere
- Speech and language delays
- Attention and focus issues
- Emotional grounding problems
- Problems with coordination and motor control
- Struggle with learning challenges (when we don’t know midline)
- Two unskilled hands (when we don’t have good laterality)
- Handwriting issues: horizontal and diagonal lines.

Gravity line (S. Goddard)
Dyscalculia (difficulty understanding numbers) 12x9 =

Task	Hemisphere
Recognize numbers/symbols	right
Find meaning of the symbols	left
Regular necessary procedure/carry out sequential tasks	left
Hold number facts in working memory	left
Find number pattern and align numbers	right
Integrate result in working memory	left
Recognition of number obtained	right
Test plausibility of answer (estimation)	right
Articulate the answer	left

- 9 changes in hemispheric dominance
- Cerebellum: sequence

Newton's law (anti-gravity mechanism)

"For every action, there is an equal and opposite reaction" (anti-gravity)

- Counteracts force of gravity
- First job of proprioceptive/vestibular system
- Triggered at birth by tactile and proprioceptive/vestibular sytem
- Triggered at birth by Moro Reflex (Babies sleep on their back, Moro kaget kagetan is activated). Genetically we are born flexion (crunching), and we activating flexion and extension as we mature.
- Mostly in extensor muscles.

Antigravity development

- Born with physiological flexion
- In prone the baby manipulates itself
- In supine the baby manipulates the environment
- Prone and supine postures mirror each other.
- Prone 3 months: baby pushes up on forearms with head up in midline
- Supine 3 months: bring its hands together in midline equal to distance in prone
- Prone 4 months: baby pushes on extended elbows
- Supine 4 months: baby extends elbows to reach for toy above body. (Mirror each other)
- Later: pushes body up (gallant and locomotion).
- When observing a movement problem on the vertical plane (standing) – trace it back to a developmental pattern that can be worked through lying on the horizontal plane (belly).
- Don’t work so hard if it is not working, take it a step back (take it horizontally, if needed go back to archetype).

Antigravity extension

- Development of vision with antigravity (as newborn can lift head up, gain vision)
- 3 months: babies cross midline with their eyes and can follow to grab a toy
- Newborn to 2 months:
  - Chest wall triangular (we were born in triangle when we need core rectangular)
  - Upper chest tight due to physiological flexion
  - Belly breather
  - Vision 2cm, cannot move between objects
- 3-5 months:
  - SCM provide stability to ribcage
  - No intercostal spacing, ribs tight
  - Extensor tone matures
  - Upper chest opens
  - Thorax begins to take a rectangular shape
  - Supine play activities abdominal connection.
  - Eyes move between 2 objects

Antigravity extension

- 5-8 months
  - Eye control improves
  - Eye body coordination develops
  - More color
- 6-12 months
  - Take bigger challenges with gravity
  - Increased shoulder stability
  - UE weight bearing and reach opens chest
  - Space for movement of upper ribs and sternum develops
  - Abdominals elongate
  - Upright postures along with rotations develop
  - Diaphragm assumes dome shapes
  - Able to judge distance
  - Throw objects with accuracy.

Grounding

- Ability to feel the ground's texture and temperature.
- Ability to feel stability of ground
- Organizes sensory and proprioceptive systems
- Brings gravity line to midline
- Activates anti-gravity mechanism
- Grounding is a sensory reflex, it tells temperature, texture. (Our feet is telling us what we are standing on
    - Toe-heel (wave in the body)

Grounding Reflex

- Activated by changes in supporting surface
- Activated by changes in texture of surface
- Search to keep feet on floor.
- Should be automatic (connected with “fear paralysis, Moro, trunk extension”).
- Keeps coordination and stability in all positions.
- Reflex:
  - Foot on table
  - Below ball of foot, and above the heel (stretch- in hook traction) elongate point of surface
  - Widen the surface: fingers go in and stretch side to side.
  - Flex a leg, knee up. Hold and support ankle, Hold right above the platela, move knee toward toe: cha.. Cha.. Cha Cha cha (the last one cascaded). Movement should come from hip.
  - Bring leg down and ground (hold ankle and flat foot it, traction). You can do both legs at the same time.
  - Grounding motion: pelvis move up and forward, it opens up the pelvis. Hold the knee, and ask them to move towards hand and you will see the pelvis come up.
  - Keep foot on the surface.

1. Grounded standing and Activation of Horizontal Perspective of Vision

- A. Grounded Standing – do a stomp or do heel-toe wave. Take a step into a different surface
- B. Grounded Standing & Activation of Horizontal plane of vision – hold stick (a line bar) in front of partner, and move it up and down. (Activating vision).
- C. Grounded standing & activation with horizontal vision Isometric –
  - Do step B
  - Hold elbow up (ask partner to push down), and move the line bar in the same direction of the body. Make sure you hold the elbow (palm on elbow).
  - Push and hold arm down (ask partner to push body up), and move the line bar up. Then do these two exercises with the line bar going opposite direction.
- Hand palm in upside down with partner (alternating palm direction) – hold and slowly move on a walk forward, backward, side to side.
  - You can put shapes, numbers, colored circle, subtraction, addition (put academics)
  - Unless you are teaching direction, don't say right or left. Just say we are moving this side, that side
  - When moving direction, use their right and left, you can also add north, south, east, west.
  - When you are doing cognitive, they can look down. But if the partner is not steady, keep their line of vision forward.
  - Add beat/metronome for them to step into the rhythm.

2. Grounded walking & horizontal visual plane anchor

- A. Grounded walking
- B. Grounded walking & horizontal visual plane – walk along a line on a wall.

3. Tactile for Grounding

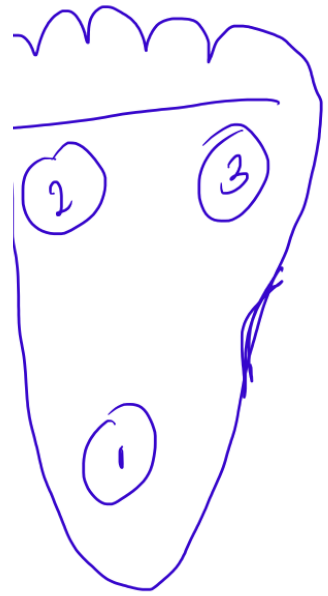
- Start with greeting the body.
- Nice rounded hands, no fingers. Embracing squeeze, take it to the table. Starting from palm up the arm length. When you get to the joint, bring hand up and not force the joint to ground the extremity.
- Greet the body again, and go to the leg, embrace the foot and up the leg.
- Wherever they are, do that with them there.
- Calming essential oils on diffuser.
- Move table not directly below the light. Turning the lights off/down is more calming.

Stability

- X-axis - horizontal stability line
- Ability to maintain firm position in space
- Ability to hold any given posture.
- Keep balance of upper/lower horizontal planes of body (our vertical brings our body horizontal plane)
- Maturity of ALL horizontal lines for sensorimotor integration
- Need 3 supporting points on each surface (tripod is more stable than two).
- Hip is most active area; shoulders balance (hip leads the way)

Gravity/Stability

- Operates simultaneously together
- Gravity reflex organizes body vertically
- Stability distributes weight on a horizontal plane
  - Feet
  - Hips
  - Clavicle/shoulder
- Gravity reflex gives awareness on one side
- Stability reflex gives awareness
- 3 points of contact form arches
  - 1-2 lateral longitudinal arch
  - 2-3 transverse arch
  - 1-3 medial longitudinal arch



Arch

Weight is directed evenly to support.

Flat - not supporting.

Alignment/stability

- Sensory receptors on bottom of foot alert brain to slightest weight shift
- Feet/ankle alignment profoundly affect leg length pelvic positioning.
- Brain always trying to maintain pelvic balance
- Muscle tone regulation needed in foot arches
- Stability compromised by less than 3 supporting contacts
  - Normal arch – neutral alignment
  - High arched print – supination-rolls to outside/lateral
  - Flatfoot print - pronator-rolls to inside/medial
- In therapy, barefoot. But there is room for orthotics (this can be done for home or in school). We do not develop our foot, our shoes makes our foundation, not ourselves so have poor stability.

Alignment/stability/grounding

- When weight is centered through gravity point, our toes spread out and interosseous muscles stretch.
- Stimulates positive supportive response
- Body senses it is on solid ground
- Whole body goes up.
- When the arch drops, knee rotate, hip inbalance

Stability/grounding

- Grounding: grounding reflex
- Lateral: babinski
- Stability: food tendon guard
- Anterior/posterior: foot grasp
- When sitting – need minimum 3 points
  - Coccyx is one part of a three part support in sitting
  - Weight is distributed between the two ischium and the coccyx, providing balance, grounding and stability

Stability

- Hips are the BOSS
  - Center that effects chain of alignment and movement
  - Facilitates posture, alignment, balance tension
  - Position effects position of spine
    - If you have difficulty in potty training, look at the pelvic
  - Dictates motion of neck, shoulder, arms
  - Activates abdominals and pelvic floor

- Group activity: sit and stand, pelvic front crouching, and slouching (try walking, and try raising up hands)

Immature horizontal planes

- Head and eyes not separate = tunnel vision (we need eyes separate from head)
- Log rolling (head moving when they are reading)
- Head turns to read causes motion sickness == Brain stops
- Brain tells eyes to open further = stare (dissociation)
- Writing very difficult = learn skill at cortex level
- No coordination of vision and auditory memory
- Neutral hip alignment:
  - Unbalanced pelvis: COG shifts
  - Distress
  - Body seeks to erect head with eyes level
  - CNS overload-Li BIC system hyperactivity
    - Horizontal spine rotation (archetype) – lots of rolling

1. Stable Standing & Activation of horizontal lines

- A. Stable standing – ask partner just standing
- B. Stable standing & activation horizontal lines – ask partner to stand and move a horizontal stick up and down (always end at up).
- C. Stable standing & activation horizontal lines isometric
  - Do B
  - Partner resist pushing on elbow and move the stick vertically down and up.
- D. Dynamic stability & auditory activation
  - Partner standing, do horizontal spine rotation
  - Watch their feet, you want them to go babinski on one but not letting go stability.
  - Give clicker logical when they turn right (activation left brain).
  - Give clicker rhythmical when they turn left (activation right brain)

2. Stable Walking and Horizontal Line Anchoring

- A. Stable Walking – invite partner to walk all over the room, experience:
  - Walk with clenched teeth
  - Walk with clutched hands
  - Walk with tongue touching upper mouth and back (a lot of CP kids do this)
- B. Stable Walking & Horizontal Line activation
  - With a tape on the wall

3. Stability – gait activation & auditory anchoring

- A. Gaits lengthening and shortening
  - 1. Upper Body
    - Stroke out from sternum to shoulder (twice on each side)
    - Put hands on both side of shoulder, ask them to “move into hands”, feel the shoulder stretched.
    - Put hands on the inside of the upper arm, and ask them to “hold” resist
    - If this is hard, then: do one side, hold shoulder and inside of arm and ask to hold. Do the other side. And then go back with hands in the inside of upper arm right and left together
  - 2. Lower body
    - Stroke from core to hips (twice on each side)
    - Put hands on hip and ask to move into hands.
    - Put hands on the inside of thigh and ask them to hold.
    - On one side of body: hold hip and inside of thigh and hold. Do other side as well. And go back with 2 hands inside the thigh and hold. It should feel stronger
- B. Gait lengthening-shortening with auditory anchoring

More stability exercises

- Use Pilates ring or a ball.
  - Inhale and exhale through teeth “csssss”
  - Hold the ball/ring in front of the body, in the middle (arms length).
  - Move the ball/ring up and middle, down and middle.
- Standing and reach out
  - Stand and spread arms side to side, wide as if trying to reach the wall.
  - Bend elbow, fingers toward front, hold on elbow.
  - Ask to resist
  - Bring arms slowly down towards body. If no partner, try to do resistance on your own
- Practice on your own:
  - Hands on hips, shoulder flat on back.
  - Then move elbow forward.
- Flamingo walk ? (Not clear)
  - Stretch arms out 90 degrees, hold elbow and hold to get trunk extension
  - Flamingo walk, traction out
  - Hold the line bar – ask for knee to touch the bar, leg up, knee Touch Bar, and out to the side

Balance sensors

- Brain
- Inner ear (semicircular canals with hair cells)
- Muscles (muscle spindle)
- Sense of touch (pacific rufini)
- Eyes (gaze stabilization - allows to move one direction and eyes can move separately than head, I can walk and my environment doesn’t move). I can jump and keep eyes on a target.
- Joints (joint kinesthetic receptor)

- We need to have mature tactile systems, we need mature vision.

Balance reflex

- Set of body and brain reaction to search equilibrium when stability is lost from displacement of gravity center
- Body-brain reaction search for body equilibrium
- One of earliest reflexes to develop in utero (the vestibular system is the very first system to develop in utero).

Balance reflex

- Controlled by vestibular system
- Responsible for static balance
- Responsible for dynamic balance (balance while moving)

- 2 Types of balance
  - Static – otoliths report on the position of the head with respect to the pull of gravity. The ability to stand and hold body still
  - Dynamic – (semi circular canal) rotational acceleration detected by semi-circular canals.

- Children with poor static balance need to move
- Can have dynamic balance with poor static balance
- Poor static balance = decreased concentrating and fidgeting.
  - Balance is the art of being still
  - You can have a child who is running at the end of a cliff and not fall.
  - Some kids with ADHD who cannot stand still they have weak vestibular system. If you keep them still, they have lost concentration.

Vestibular system

- One of the first system to develop in utero
- Equilibrium, sensation of motion and spatial orientation (telling us where we are oriented in space, where my body is).
- Allows for upright posture
- Stabilize the eyes during head movement
- Recognizes direction of gravitational force.
- Organizes postural control
- Attention (weak vestibular system, bad attention span)
- Being secure in your environment (clinging to wall or grabbing someone, or the ones who try to skip their turn to do a skill).
- The most protected part of your body (it is so small, a tip of your pinkie finger imbedded deep in your bone).

Anatomy/function of ear

- Auditory/hearing: transmit sound vibrations to cochlea
- Balance: structures of the body labyrinth (you can balance and not be able to hear in process, the listening ear and the body ear)
- Major source of sensory energy, picking up vibrations at different frequency
- Protects us from sounds we do not want to hear
- Body-brain connection that tells larynx what to do (the vibration and brain activity that tells the larynx what to do: you can hear a pitch and then you can imitate it). If someone is tone deaf or cannot speaks, there are the motor area that effects the ability to sing or speak.
- Gateway to language center of brain (with Down syndrome: they often have small ear canal. Low tone and mouth motor system that kept them from talking right). Small ear canal can lead to speech delay

Vestibular system/Trunk extension Theresa's note

- Vestibular system turns on auditory processing
- If hearing isn't connected with vestibular system it is not fully integrated in auditory cortex
- If need to use eyes with ears vestibular system is not turned on (you should be able to close eyes and localized sound where it is coming from and follow it).
- Auditory system should improve with trunk extension
- Need integration/myelination with trunk extension then auditory discrimination will develop
- If not integrated with trunk extension then hearing will deteriorate quicker (as you age, and not having trunk extension, hearing will deteriorate).

Sound is movement

- Auditory and vestibular systems have same function: perceive movements
- Vestibular system: perceives slower movements that are seen and felt
- Auditory system: perceives faster, oscillatory movements that are heard
- Auditory and vestibular systems both use excitation of hair cells
- (The lower frequency you feel it. The humming bird, the occilation is so fast, all you is tweets)

Hearing vs listening

- Hearing: passive ability to register sound
- Listening: active ability to turn in on sound messages and tune out sound (being able to process the information that you hear)
  - Auditory processing
  - Auditory discrimination
  - Attention span
  - (Trunk extension and gravity reflex)

- Kids don’t listen because they don’t move anymore. They are not ready for school anymore. Head down sitting down: auditory closed (when they are on screen)

Vestibular problems

- Degree of clumsiness
- Feeling “off”
- Could take picture
  - Concentration
  - Out of focus
  - Not centered

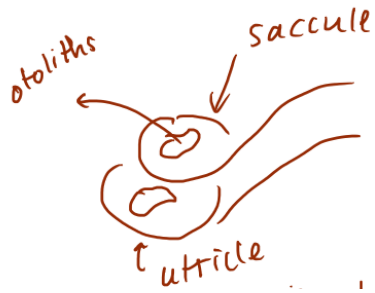
Auditory problems

- Academics
  - Correct aim in time and space at relevant information
    - Focus
    - Attention (They hear sound but cannot make sense of it, and if vestibular is off they are so busy keeping their body up)
    - Note that vestibulocochlear nerve is located near brain stem

- When doing visual work with the kids (with the line bar) it is important to have a wall in the background. Solid clear background and not having the line bar in space (or busy background).

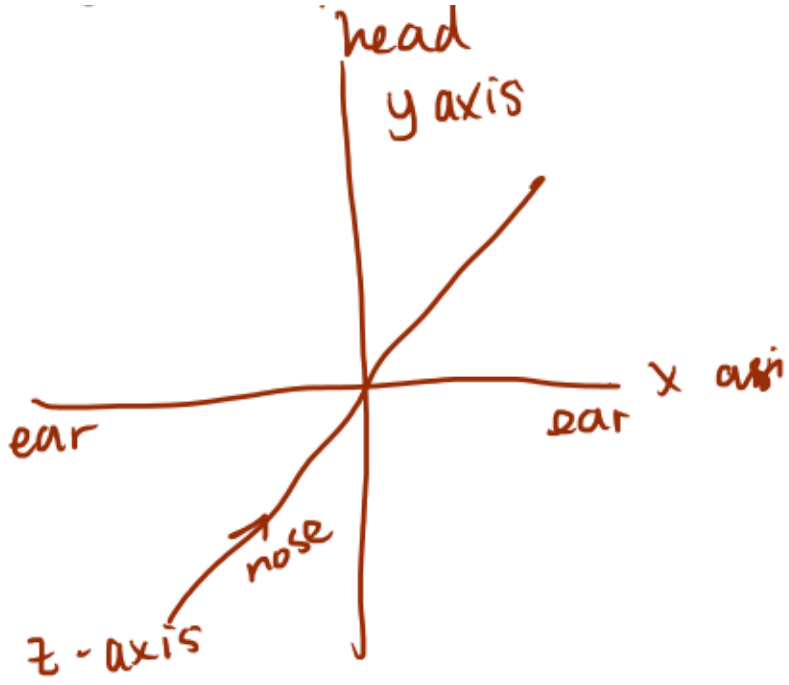
Vestibular system

- Two main structures
  - Semicircular canals
    - Rotation (dynamic balance) – 6 degrees of freedom: 3 rotations
    - represent 3 spatial planes of body (3 dimensional axis x,y.z). We are able to move in the 3 planes of dimension because the vestibular system is driving it.
  - Otoliths in Utricle and Saccule (little stones)
    - Translational /linear movement – 6 degrees of freedom: 3 linear / translational movement
    - Linear acceleration (static balance)
    - Detect gravity
    - Upside down or right side up



Acoustic reflex

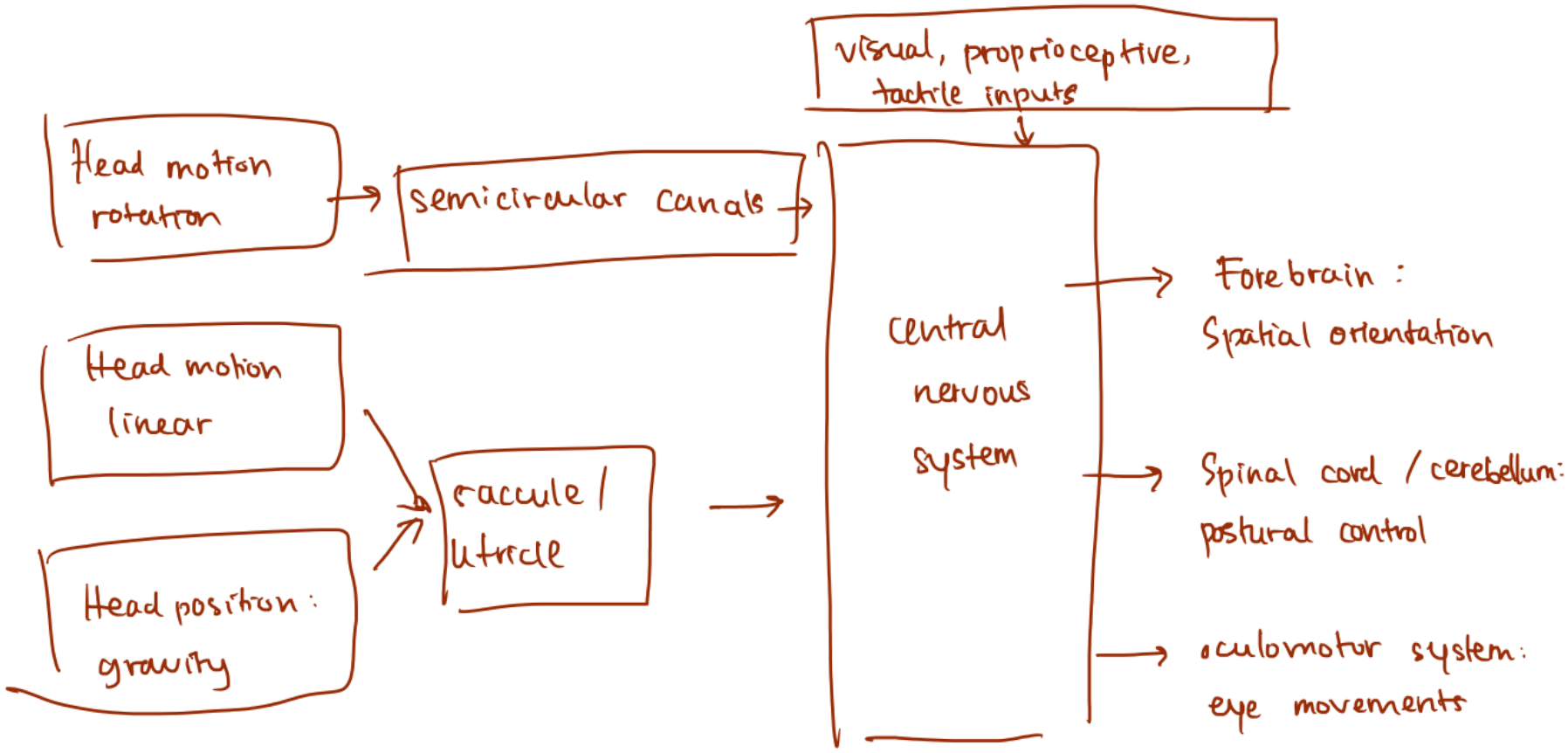
- Function: temper the force of sound vibrations that reach the cochlea
- Spatial orientation: xyz axis



Postural sway

- The upside down triangle – significantly

Vestibular system functions



vestibular weakness symptoms

- Dizziness when reading or focusing on an object
- Gets motion sickness easily
- Spin and spin without getting sick
- Bumps into things (clumsy)
- Difficulty doing math work
- Misdiagnosis of learning disabilities, especially dyslexia, dyscalculia and other reading disorder
- Spatial awareness weakness
- Sensory issues
- ADHD

1. Kinesthetic orientation in space & anchoring of auditory orientation in space

- A. Kinesthetic orientation in space
  - Close eyes, tell we are going to turn 90 degrees to the right and stop, squeeze. Continue until they come back to 360 degrees.
  - Ask them keep eyes closed and turn 90 degrees on their own
  - Variant: do the same, but ask them to open eyes after every turn
- B. Kinesthetic with trunk extension
  - Do A but after each turn, ask them to do trunk extension. Come up on tip toe and then heel.
- C. Kinesthetic & auditory orientation
  - Do like A, but after each turn give sound stimulus:
    - Right (digital logic stimulus)
    - Left (rhythmic stimulus)
    - Or put music in midline

2. Circles-rotation of the body and anchoring with stationery visual object

- A. Circles-rotation with whole body
  - Partner to stand straight, feet together only slightly separated.
  - Hold to stabilize at bottom of ankle, hold the shoulder from behind and rotate to the right (make full circle rotation) 3 times, and then opposite way: left rotation 3 times. Check motion on the core, if they flop, then show them midline (hold sternum and core, bring the body up).
  - Do it with stabilization below the knees
  - Stabilize the hip
  - Stabilize the shoulder: hold the the shoulder with both elbows and upper room and fingers on the temporal making a crown, rotate the head.
  - Notes:
    - You can also do this on the table: joint rotation on wrist, elbow, shoulder, all other joints. These are all feeding to the vestibular system (a lot of input)
    - Some people get drowsy just by watching the movement (weak vestibular). Some people can rotate forward but jerky especially when going backward. Do only 2 rotation on people with weak vestibular
    - Also start with the shoulder rotation if they have weak vestibular.
    - When handling children alone: you are sitting, have then stand in front of you, facing same direction as you. Anchor their feet with your feet, and move their shoulder, waist, hip, knee.

2. Circles-rotation of the body and anchoring with stationery visual object

- B. Circles-rotation of whole body.
  - Do as A
  - Ask them to concentrate on a stationary object behind them (touch circle on the wall).
    - If you are behind them, say green, and when they turn again, they have to say the shape (triangle). When they turn right, say 2+2, then ask them to turn left and say the answer.
    - You can add metronome so that it goes with the bead.
    - You can say 6, and then they have to pick up the equation.
    - Use pendulum ball (dog ball) on a string hanging from the classroom ceiling. Hold the ball stationary at first then have them hit the ball
    - Use ball with alphabets on them, or numbers. Point the ball with finger, and wherever the finger lands, you have to read it the letter/number. If number: add one more, if alphabet: you need to say a word begins with that letter.
    - You can also have the pendulum ball go to right hand and left hand.
    - Rotate and see cards with number unit on them. Do this one first (flash cards with the quantity and the number on it
    - After the number unit cards, you can ask them to say what numbers on your hand.
    - You can also do math operations: right (equation) and left (the answer)
    - You can use a balance board, stand forward, turn left pick up letter, turn right put letter on magnet board (before doing this, make sure they are comfortable with the balance board).
    - Kirchner Chart (the arrows), print 2 rows at a time on a big paper, ask them to look at the arrow, say the direction, look in the direction of the arrow with their eyes (without moving head). Then ask them do follow that direction with their body. (Jump right left, down, and jump up).

3. Game: sound orientation in space

- A. Partner sitting down with eyes closed or after a turn to right/left
  - If you cannot localized sound, you cannot hear
  - Clap behind, clap in front side ways, and have them point to the direction of the sound.
- B. Standing up, close eyes
  - Turn 90 degrees, and stand up on toes, trunk extension
  - And give the clicker sound (right ear: C, A, T, left ear: CAT)
  - OR do: stand them on a stool (improve sensory awareness), turn right: see word, turn left: see word. Teacher do a metronome beat. (Right input short term memory, left input long term memory). You can do sight words on orange background. And we can go with math: right equation, left see answer, and then we can put a blank and they have to say the answer.

Eye movements

- Optokinetic – head is stationary. Object moves
- Vestibulo-ocular –vestibular input head moves
- Smooth pursuit – following moving object
- Saccades - acquires new object (this is used for reading)
- vergence movement – eyes move inward or outward.

- Partner on the floor, the ball above them and then move ball around to follow
- Vestibulo-ocular reflex: eye move head move opposite direction
- Optokinetic reflex: head stationary eyes follow object, developed at 6 months of age.
- Nystagmus – involuntary repetitive movement of eyes.

- Fixation: period where the eyes are locked towards an object
- Saccades: eye movements between fixation
- Visual span: how many words we can read before and and after the currently fixated word.
- Smooth pursuit: fixate constantly on tracked object, and there won't be any obvious saccades.
  - Reading comprehension issue or reading issue
- Eyes do not move continuously along a line of text
- Make short, rapid movements (saccedes) intermingled with short stops (fixations)
- When reading eyes don't travel smoothly, lock our eyes towers every third or fourth word
- Trained readers have higher visual span, allowing them to cover more text with less fixations (3-5th grade)
- Fixations and saccedes are excellent measures a visual span.

Binocular coordination

- Children have to learn eye teaming
- Often undetected in vision exams
- Nothing to compare to difficulty reading, writing, math

- Cover right eye
- Cover left eye
- Read with both eyes: more difficult.

Saccadic eye movements

- Ability to move eyes quickly from one focus to another
- Important for reading skills
- Indicative of under developed cognitive processes
- Contributes to other learning disabilities.

- Hold 17” from eyes
- Start with eyes in center
- Look at owl then center
- Look at fish then center
- Change order randomly

- Perform 15 seconds (everyday – 2-3 times, rest in between do some motor activities. They should already have good convergence and divergence, these are very simple and cannot go wrong with it). Research: eye visual therapy. Without glasses on
- Head should not move
- One sharp eye movement (several times a day) give this to aide

- Optokinetic reflex
  - Hold your finger a few inches in front of you and move fingers in fast motion
- Vestibular ocular reflex
  - Hold fingers stead, move the head side to side

Head Righting Reflex

- Reaction of head motion towards the whole body position and movement as a result to spatial/structural changes
- Involves complex muscular movements in response to a stimulus
- Body's reflexive attempt to keep the eyes level against the horizon
- Rely on an aligned pelvis, a springy ribcage and a stable, flexible cervical spine (get these from archetype)

Head Righting

- Ocular head righting
  - Visual information detects body displacement
  - Eyes keep focus and stability
- Labyrinthine Head righting
  - Eyes closed
  - Information from vestibular and proprioceptive system headrights

- Head righting requires maturity of visual and auditory system
- Combination of visual system inputs, vestibular inputs and proprioceptive inputs make postural adjustments.
- Good muscle tone in neck muscles allows movement in jaw and cranial bones for head righting. (Level eye and level jaw are going to give us a better information to head righting, if not level then head righting not going to happen
- Temporal, sphenoid and all cranial bones need mobility to reach vestibular system

Weak head righting

- Child struggles listening to teacher
- Underdeveloped auditory processing
- Poor handwriting
- Trouble with gross and fine motor skills
- Child slouches in chair
- Laying on desk
- Struggle to lift their head to view the chalkboard
- Imperative for eye tracking when looking away from something and then looking back
- Struggles tracking words on page, skipping lines. (Take ruler or hand to follow the line).

1. Head Righting

- Exercise p. 53
- Ocular vestibular head righting
  - Sitting down in crisis cross, look at hip in neutral
  - Position the hip and support the hip with shin, give input shin up and down. Set her up for success
  - Support pelvis and pay attention to hip movement
  - Same handhold for shifting gravity line, hand on shoulder, and shift to side, squeeze and middle. Over to the side, squeeze and middle. Forward, squeeze then middle, back squeeze and middle.
  - Keep hips active (keep rectangular shape) – if you W sitting, no active hip.
  - Move to side even more, hold the hip with shin/toe or help lift the hip to keep active.
  - For Active/isometric: ask partner to hold and give your hand support.
  - If the head doesn't come right, give a little hand and say Find my hand until head come right, and hold on the side
  - Do the ocular – with eyes closed and eyes open

Head Righting p.58

- Incline: 15, 30, 45 degrees passive
- Add isometric: hold and move into my hands
- Eyes open and eyes closed
- Add in from archetype: add in shortening and lengthening in the midline (to get hip mobility). And then sway side to side, gravity line sway, and you are giving them a lot of midline input.
- Notes:
    - If head tilt with body, then with one on shoulder (hold) and touch temple slightly and ask to move into hand on temple. Head should come right.
    - When moving body forward, hold on back of head. When moving body backward, hold on forehead and ask the head to move into hand.
    - To get head righting, you can stick a horizontal tape on the wall and ask partner to level their eyes with the line
    - If partner cannot sit criss cross, partner can sit on a chair but make sure foot is supported (grounded).
    - If you are working on gravity line and not see head righting -> put horizontal line on a wall and then put the isometric (push into my hand) and you will get head righting. If you don't have it, bring back to the table and do archetype lateral flexion and extension
    - Balance board is only to be used to get them adjusted moving side to side. But not for head righting. They are just using their body weight not hip movement. You really need to start on the floor.
    - But once they have it, you can then use the balance board
        - Forward/right -> digital -> 1+1 / C,A,T
        - Back/Left -> gestalt -> 2 / CAT
    - When sitting down, you can also do rotation on hips and pelvis (stability), work on shortening lengthening: hold on sternum and core, head on back, toe on sacrum – you can feel the head righting
    - For auditory activation:
        - Forward/right -> clicker logical
        - Back/left -> rhythmical
        - After a while, you can start clicking on an angle, or put city traffic sounds in the background. Ask partner to close eyes and point out where the click location is. Change volumes of the background sound to keep testing.
    - Video: (a boy mid kindy, not talking – they were ready to label him autistic) -> repattern STNR, crawling, STNR, trunk extension, rotation, hand supporting trunk extension, use wooden box (standing inside), rotate and touch the wall behind.

Proprioception and Motor development

- Movement is a complex brain activity
- Requires coordinated response of entire nervous system
- Influenced by proprioceptive system in partnership with vestibular system
- proprioception connects brain with muscles
- Proprioception organizes postural control
- Want gravity, antigravity to connect with visual functions (as babies learn to develop, they develop their vision)

Postural control

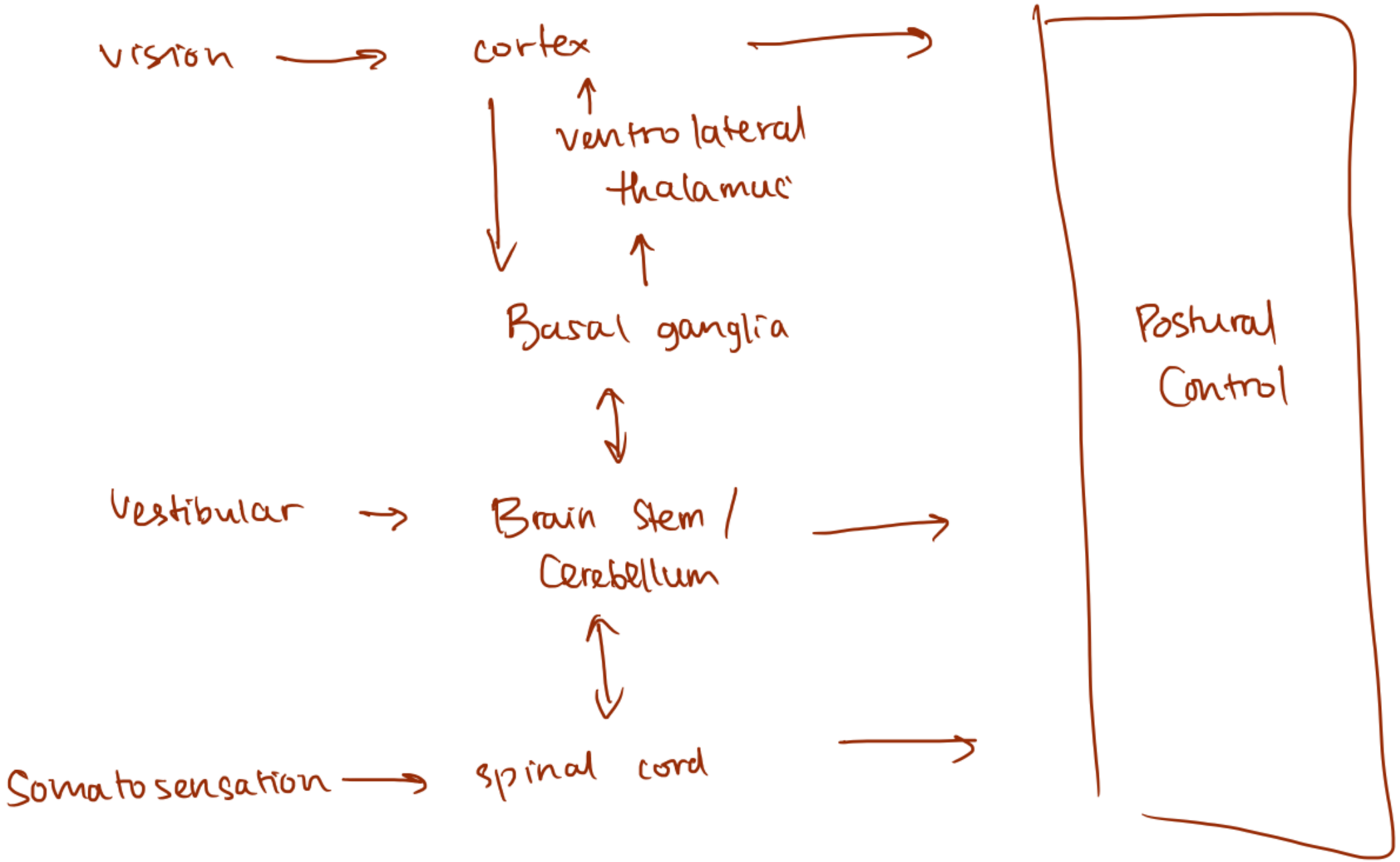
- Provides orientation and balance using 3 senses
  - Somatosensation: input from tactile and proprioception
  - Vision: movement cues/upright response
  - Vestibular: head position relative to gravity

3 levels of somatic motor activity

- Highest level
  - Cerebral cortex
  - Generates movement foal and gives command and strategy
- Middle level – this is where MNRI is working (making adjustments)
  - Basal ganglia, cerebellum, brain stem motor centers
  - Supervise the motor plan
  - Regulate activity in descending motor pathways.
- Lowest level (physical therapy usually works here)
  - Lower motor neurons, skeletal muscles
  - Execute motor movement and supply sensory feedback

- Posture:
  - Biomechanical alignment of the body

- Postural stability:


Vision
cortex
ventro lateral thalamus
Basal ganglia
Postural Control
Vestibular
Brain Stem / Cerebellum
Somatosensation
spinal cord

Posture / Whole Body listening

Inner Ear
↓
high frequencies
↓
listening / sound

Vestibular system
↓
low frequencies
↓
detect movement

↓ environment
↓
reference point in space.

↓ body
↓
relationship to gravity

Posture / Whole Body Listening

Inner Ear

(Body ear)
Vestibular system
(postural tone
movement strategies)

Limbic System
emotional control
Memory
Basic driver

Reticular formation
Arousal
Alertness
Attention

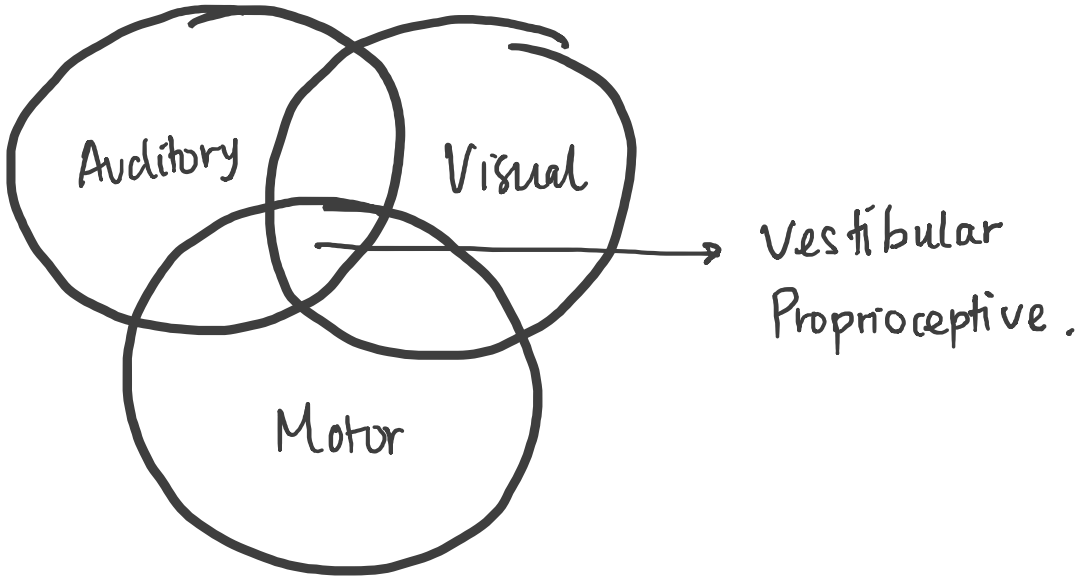
Listening ear <-> body ear

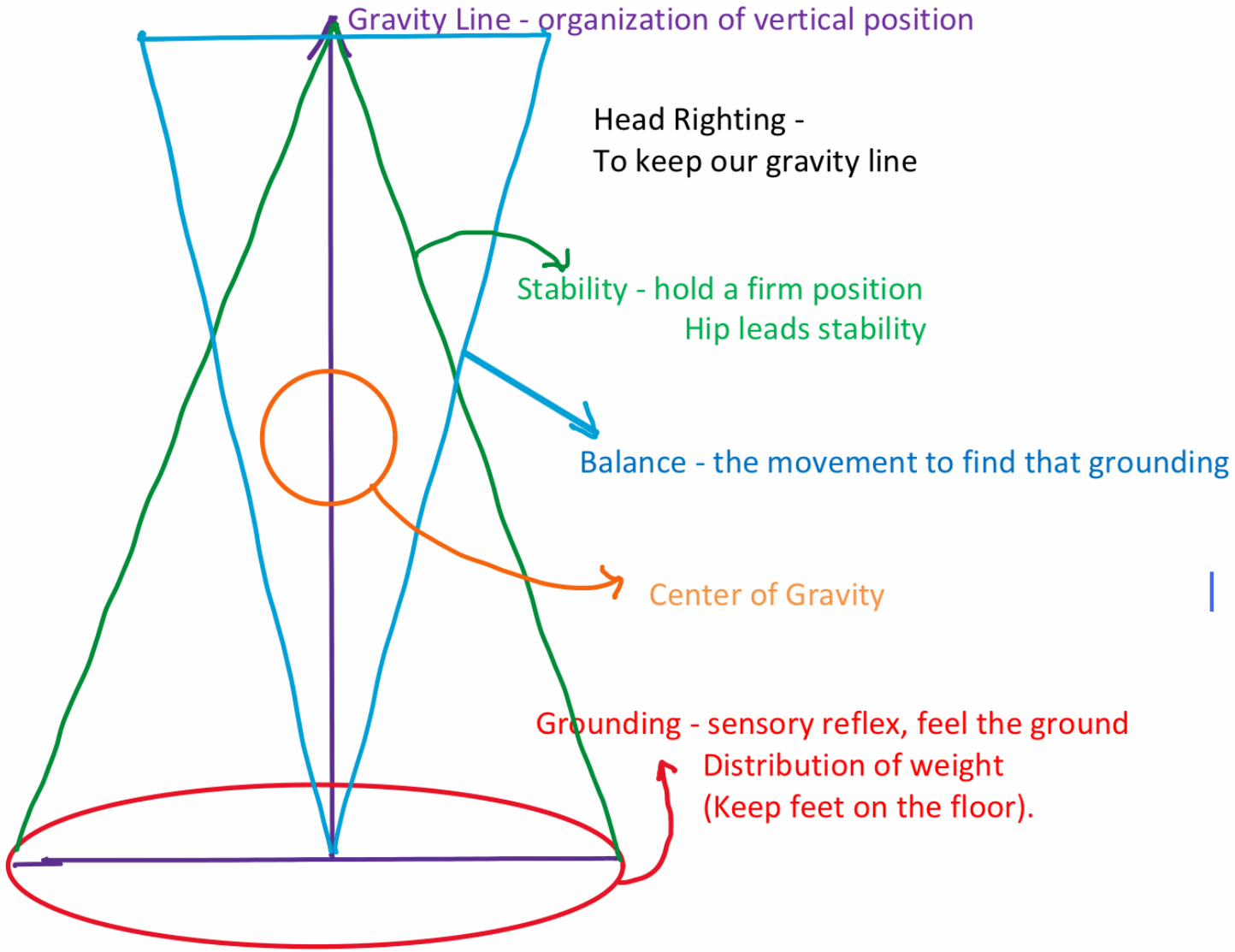
- De Quirres, Jean Ayres, Stephen Levinson: dialogue between body & auditory ears – provides gravity, stability and grounding balance.
- Sound-movement harmony:
  - Music reinforces the “dialogue” between body ear and auditory ear
- Synchronicity between movements and sounds (body ear/auditory ear) is a prerequisite for acquisition of language and academic learning
  - Nursery rhymes
  - Children’s songs
  - Music and preschools
- Do rhythm and jump (clap.. Clap.. Clapclapclap), 1,2,3 right, 4,5,6 left, 7,8 forward, 9,10 backward.

Needs of learners

- Orientation to the world = posture – in prop cog: find posture, hip mobility, help them be comfortable in their body.
- Move around environment = locomotion
- Interact with objects in environment = manipulation

Learning Challenges




Gravity Line - organization of vertical position
Head Righting -
To keep our gravity line
Stability - hold a firm position
Hip leads stability
Balance - the movement to find that grounding
Center of Gravity
Grounding - sensory reflex, feel the ground
Distribution of weight
(Keep feet on the floor).